

Non-Mendelian Genetics I (Epigenetics)

Readings: Ch12 418-423 [section 12.4]

- Epigenetic regulation during development
 - X-inactivation
 - Genomic Imprinting
 - Imprinting and disease

Epigenetics: Sometimes, genotype $\neq$ phenotype

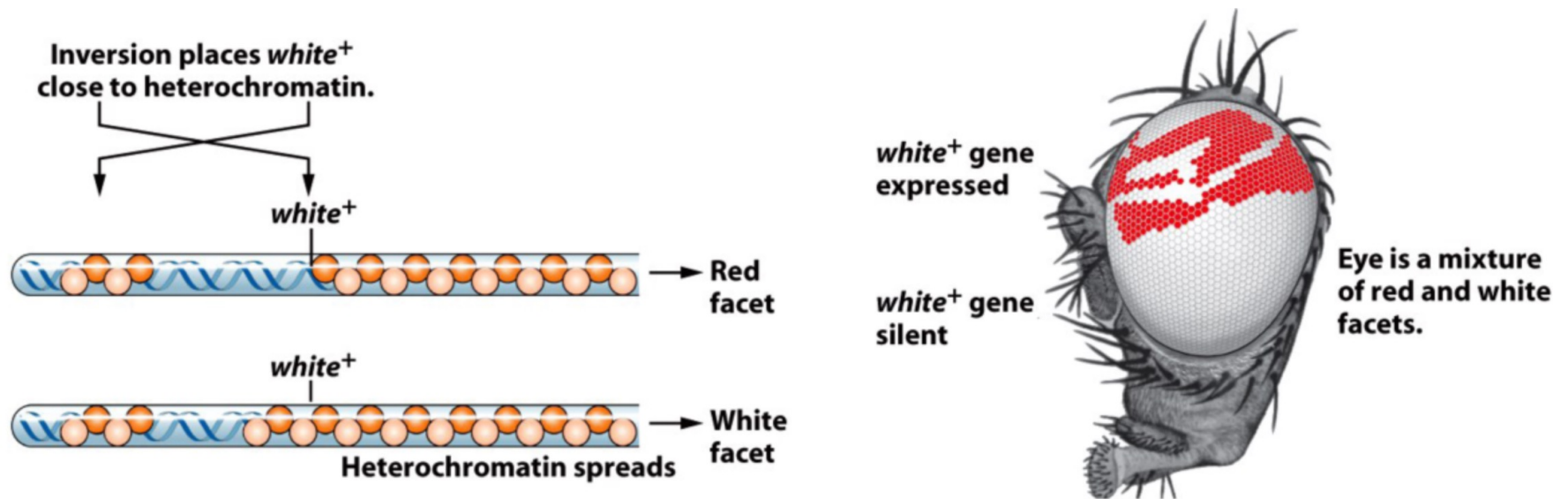


Fig 12-22

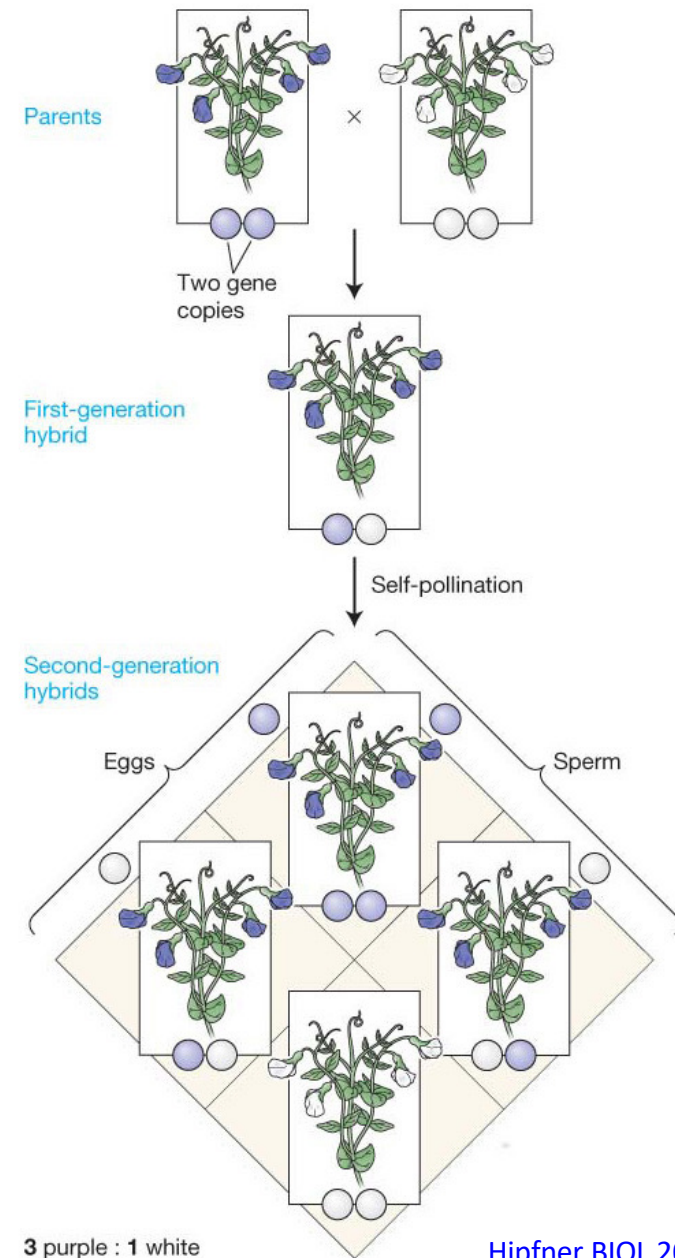
Position-effect variegation

Mendel's Laws

- 1) Each individual has two alleles for each gene, passes on only one to offspring at random
- 2) Each allele has an equal chance of being inherited (independent assortment).

Non-Mendelian genetics:

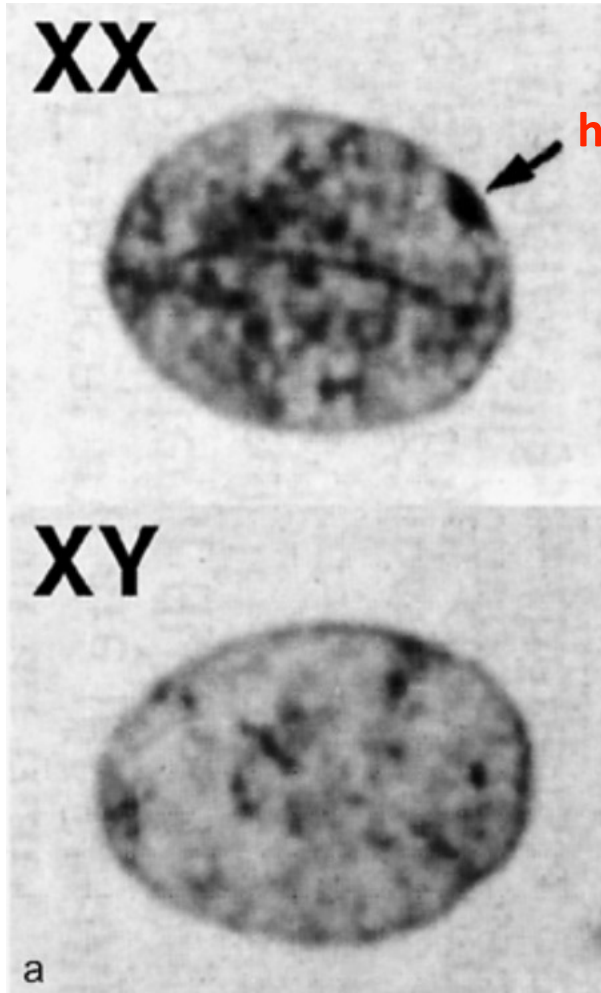
Pattern of inheritance in which traits (phenotypes) do not segregate in accordance with Mendel's laws. (Mendel's laws cannot explain the phenotype)



Q. A woman inherited an X-linked recessive mutation that caused her father to have liver failure. What is the probability that she develops liver failure?

- A. 100%
- B. 0%
- C. 50%
- D. Not enough information to predict

Dosage compensation by X-chromosome inactivation

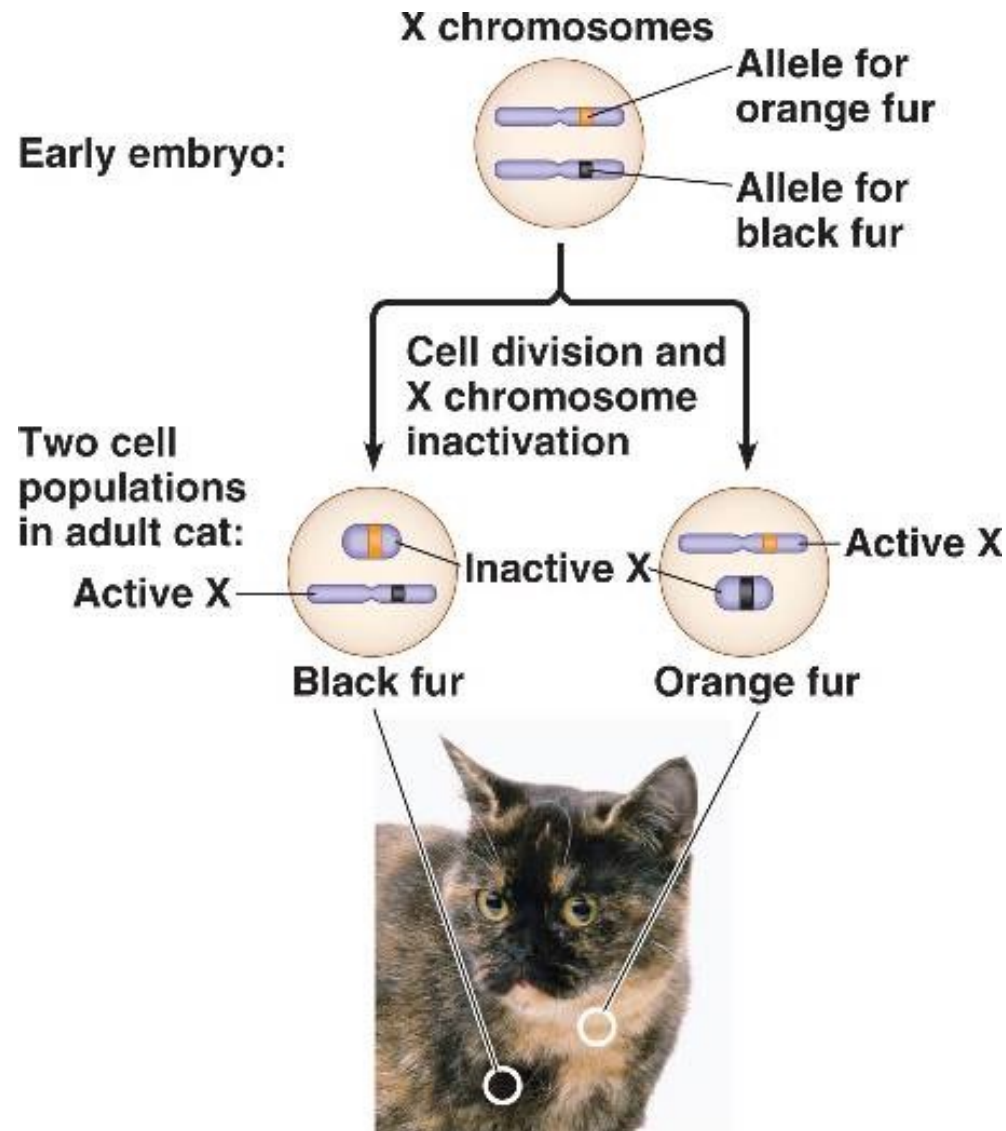


heterochromatic X chromosome (Barr body)

- “Dosage compensation” - so that males and females have equal expression levels of the ~1000 genes located on the X chromosome
- In early female mammalian embryo (~8-32 cell stage), each cell randomly “inactivates” one of the two X-chromosomes - forms heterochromatic “Barr body”
- The same X-chromosome is inactive in all descendants of a given cell - “mitotic” epigenetic inheritance

- this “mosaicism” of females can lead to unexpected genetic consequences....

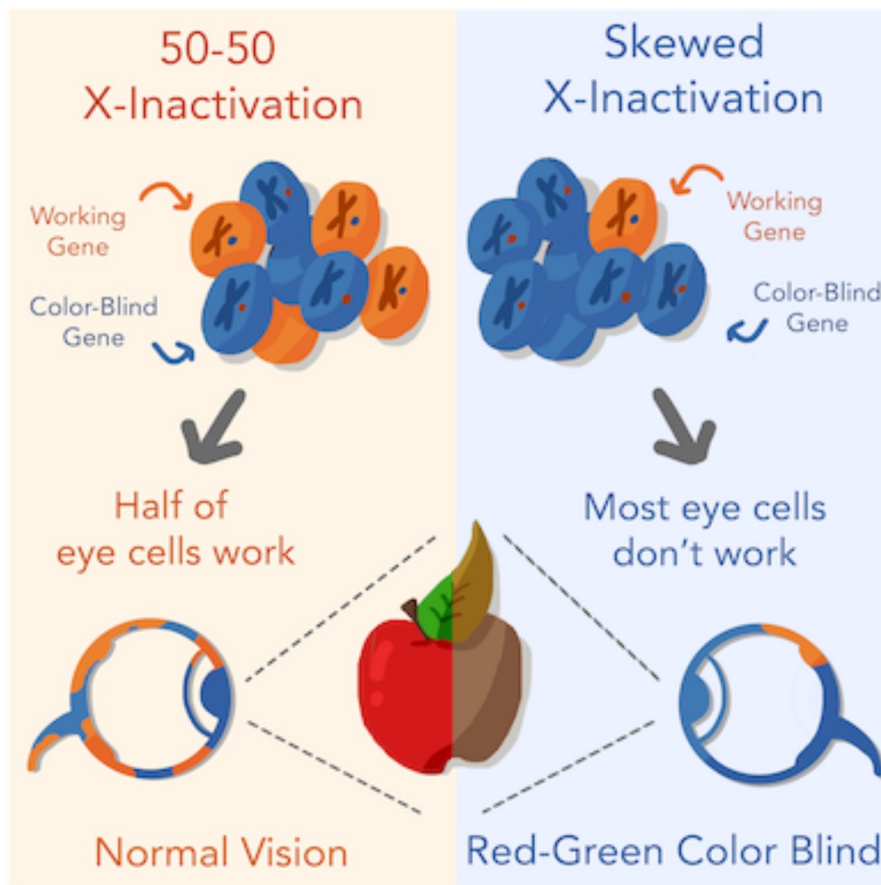
Non-Mendelian inheritance of coat colour in tortoiseshell cats due to X-chromosome inactivation



Question: which allele is dominant - black fur or orange fur?

Non-Mendelian inheritance of some X-linked diseases

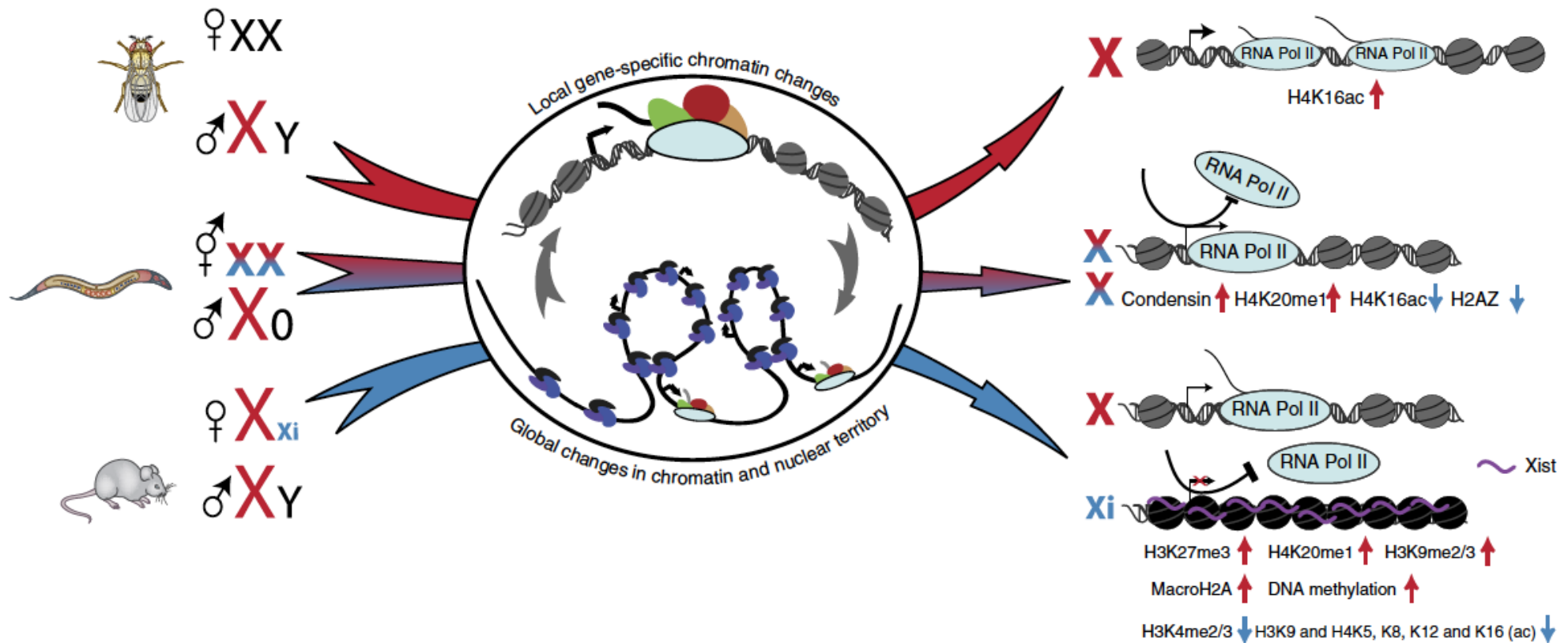
- red-green colour blindness is often due to mutation of the OPN1-LW gene on X
- mut/Y males are colour blind, but mut/+ females are sometimes as well...



- for many X-linked mutations, female carriers can display “partial” symptoms, with variable penetrance and expressivity due to “skewed X inactivation”
- some X-linked diseases like Rett syndrome are only seen in females.... because mut/Y males are not viable

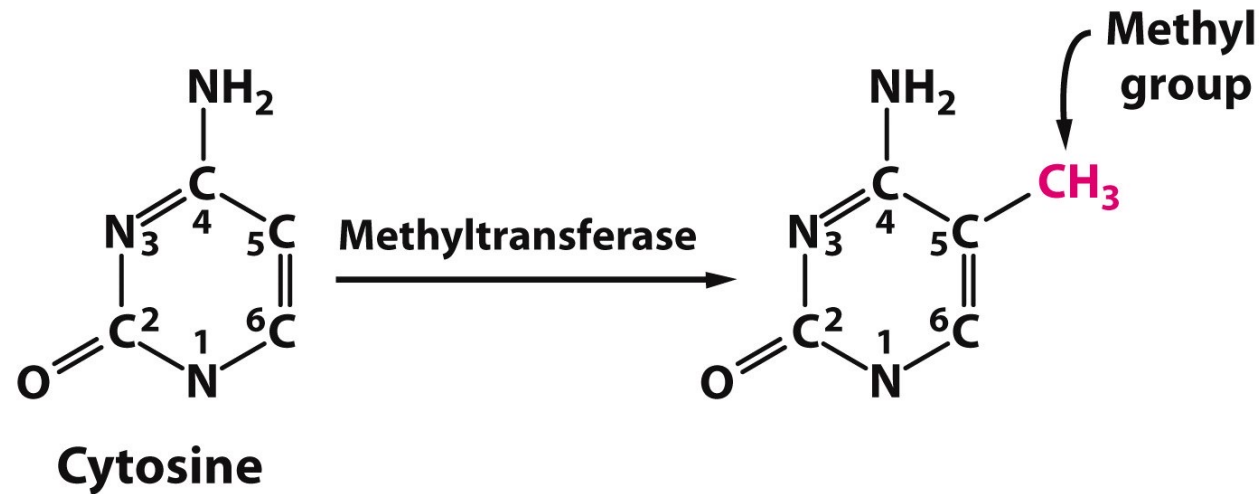
The inactivated X in humans is silenced by repressive histone and DNA marks

The difference in the number of X chromosomes between males and females is compensated at the level of transcription in different organisms.



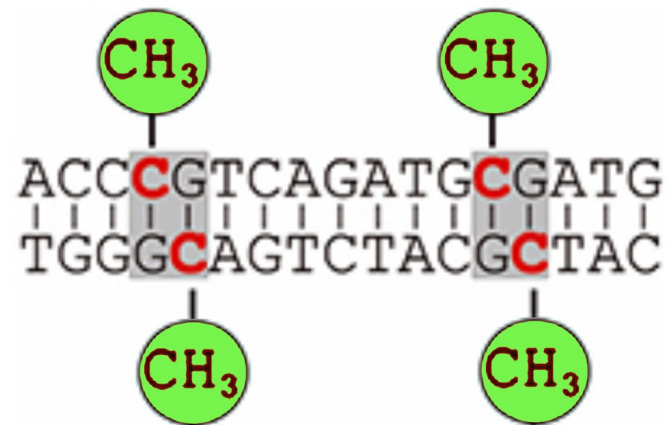
DNA methylation

- catalyzed by **DNA methyltransferase (DNMT)** enzymes



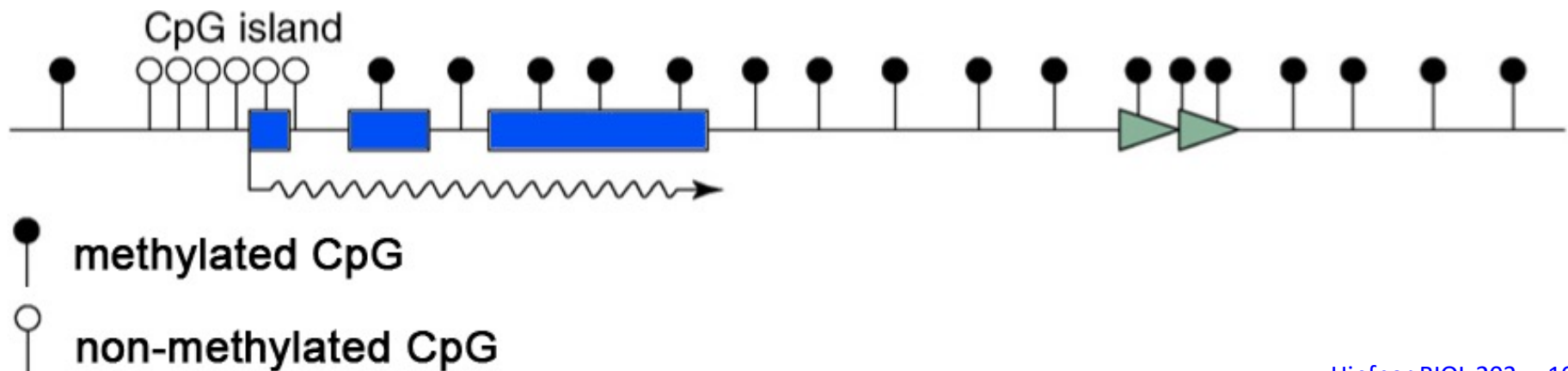
Unnumbered 12 p431
Introduction to Genetic Analysis, Tenth Edition
© 2012 W. H. Freeman and Company

- occurs primarily on cytosine in **CpG** dinucleotide (^MCpG)
- 60 ~80% of CpG are methylated genome-wide in vertebrate.

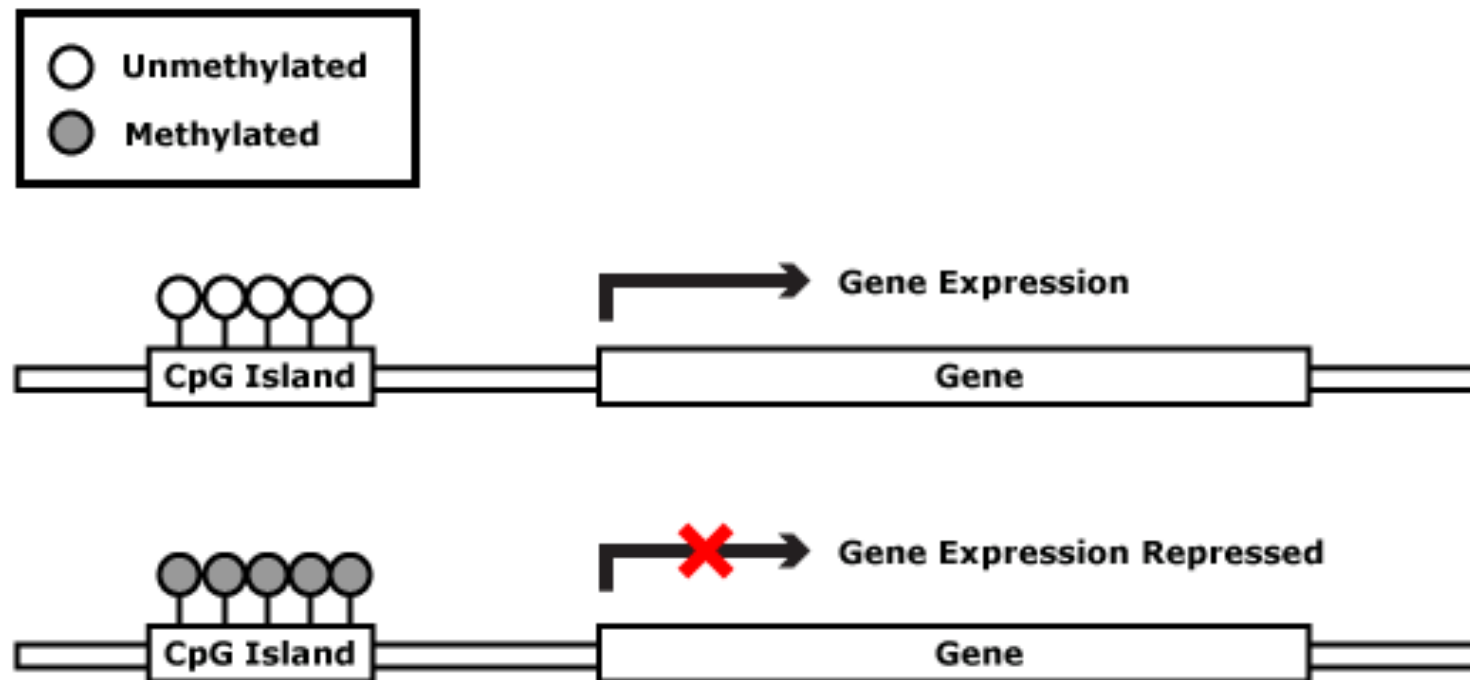


CpG methylation is not randomly distributed

- CpG methylation
 - mostly associated with intergenic regions
 - correlated with repressed chromatin state (heterochromatin)
- CpG islands – CpG-rich clusters located near promoters (60% of genes)
 - mostly (>95%) not methylated and transcriptionally active

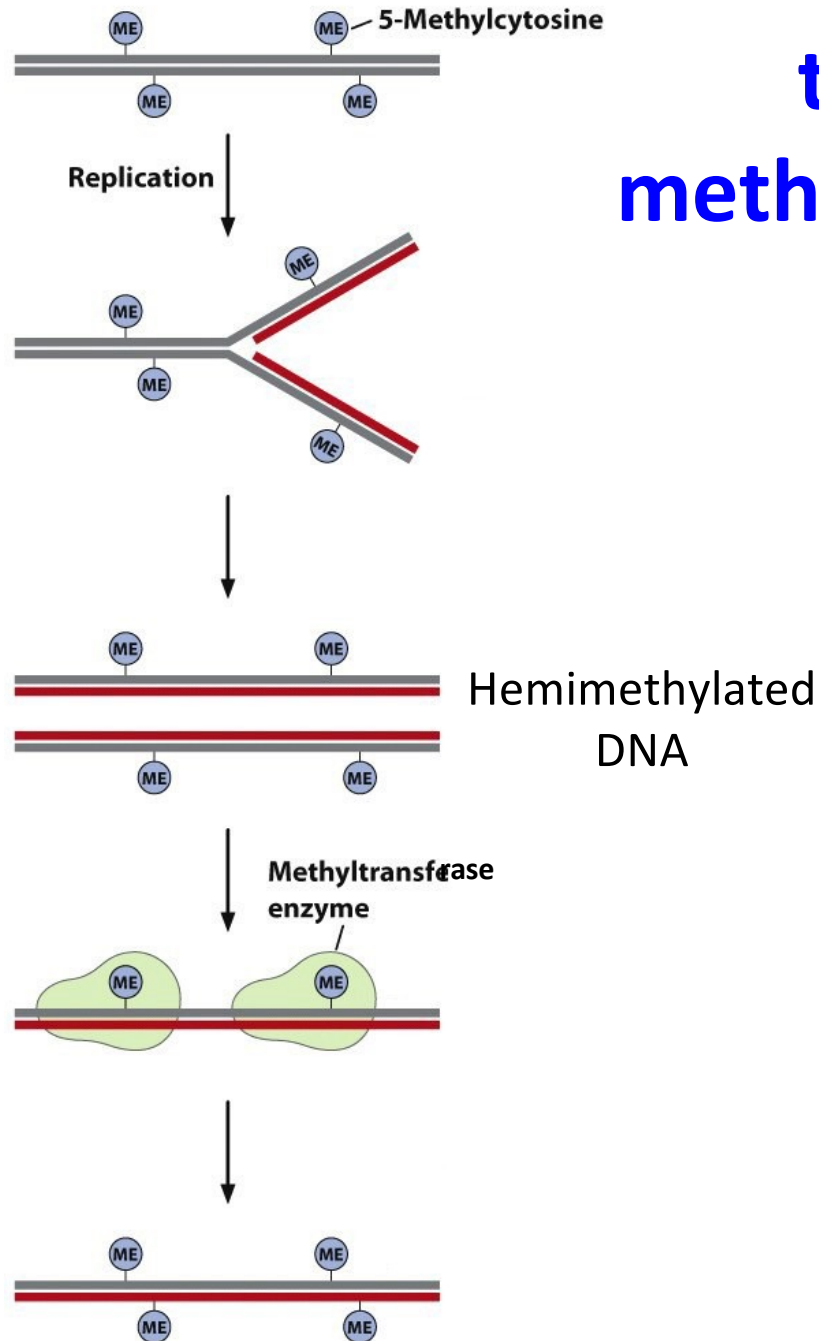


Methylation of CpG islands is associated with repression/silencing of gene expression



- Can be direct effect: DNA methylation blocks transcription factor binding
- Can also be indirect effect, due to recruitment of HDACs and HMTs that lead to repressive histone modifications

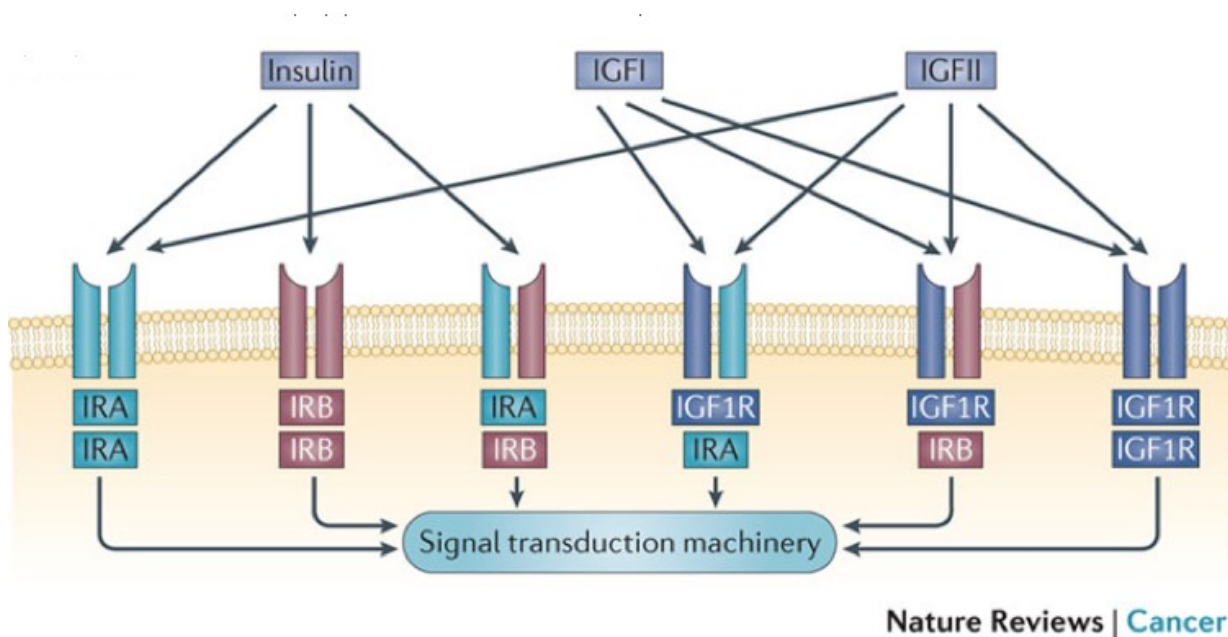
Like DNA sequences and histone tail modifications, DNA methylation patterns are passed on through mitosis



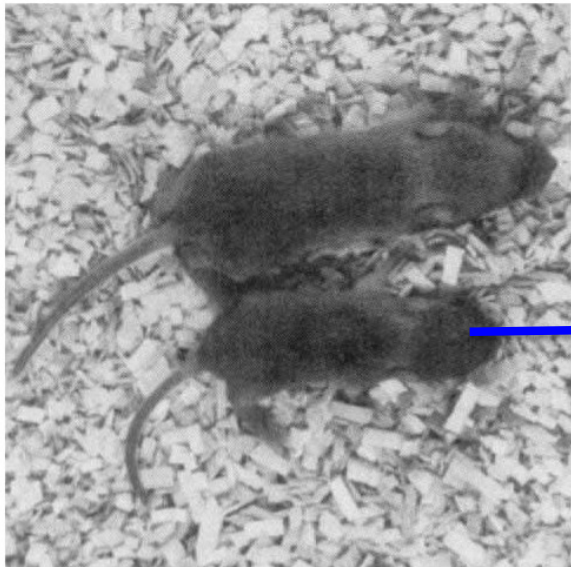
- DNMTs have high affinity for **hemimethylated** sites
 - guided by methylation pattern on parental strand

HERITABLE!

Unusual pattern of inheritance of the *Igf2* gene involved in embryonic growth



- “Insulin-like growth factor 2”
- Member of insulin gene family
- Binds to receptor on cell surface, stimulates cells to grow
- autosomal gene



Mutation of *Igf2* gene in mice – heterozygous pups born small, ***but only if mutant allele was inherited from father***

DeChiara TM... & Robertson EJ. *Nature* (1990)

Think break...

What can it mean if it doesn't matter to the fate of the offspring if, say, the paternal allele of an autosomal gene is functional or not?

Genomic imprinting

- up to ~200 genes in human and mouse genomes where only the paternal or maternal copy (but not both) is expressed – as if there were only one copy of the gene in the cell

“monoallelic inheritance”

- Which allele gets expressed depends only on biological sex of parent from which it came - **sex-specific gene silencing**
- The **non-expressed allele is said to be IMPRINTED**
- imprinted copy is inactivated by a mechanism involving DNA methylation in the maternal or paternal germline
- methylation imprint is maintained throughout life of the progeny (in somatic cells)

Epigenetic gene regulation through DNA methylation: Genomic imprinting

♀ **Maternal allele**



♂ **Paternal allele**



11th Edition Fig 12-27

In mice (and humans), *Igf2* and *H19* genes are adjacent

***Igf2* gene: maternally imprinted**

*only the Igf2 allele inherited from the father is expressed
maternal copy is imprinted and inactive*

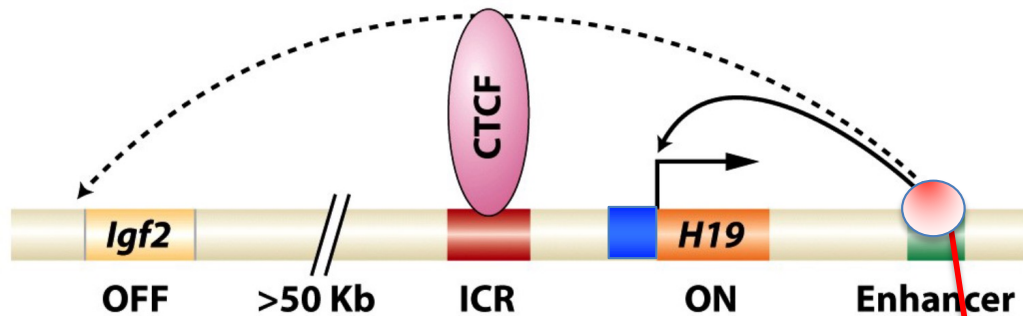
***H19* gene: paternally imprinted**

*only the H19 allele inherited from the mother is expressed
paternal copy is imprinted and inactive*

Red box: Imprinting Control Region (ICR)
Green box: Enhancer for both *Igf2* and *H19*

Alternative methylation at an imprinting control region determines gene expression

♀ **Maternal allele**



♂ **Paternal allele**

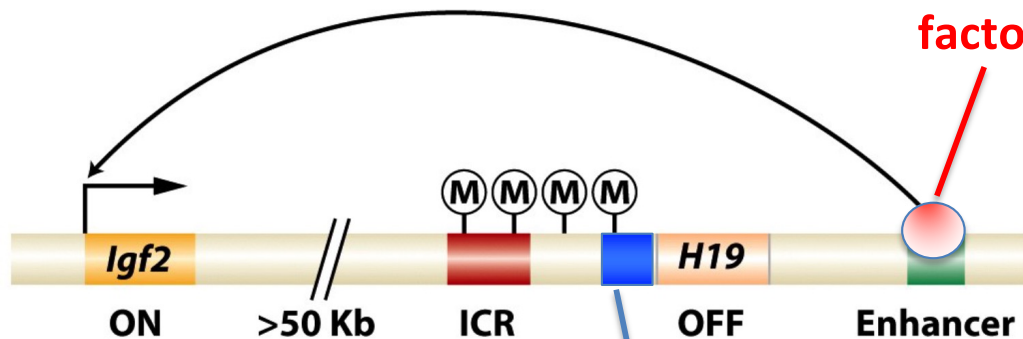


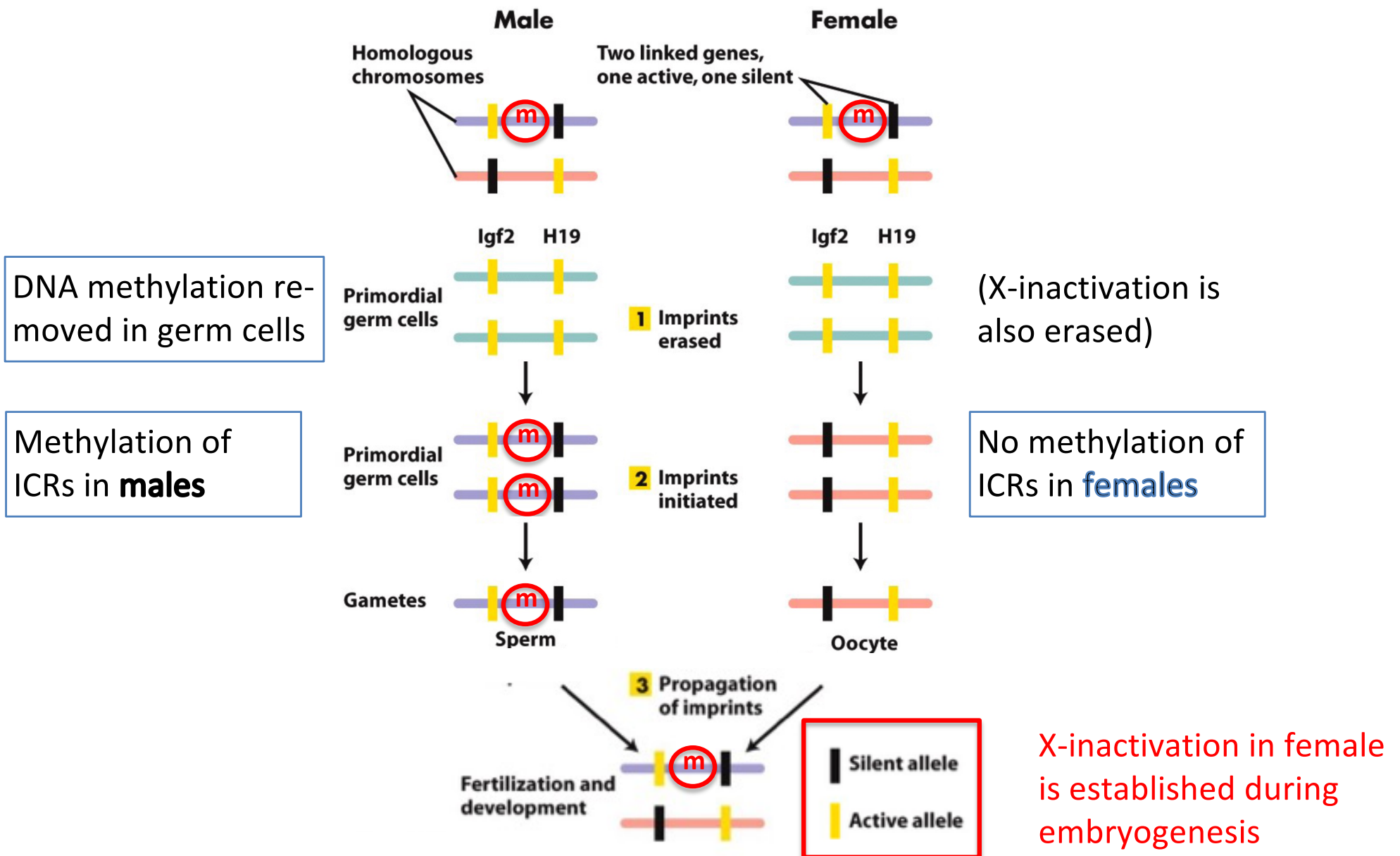
Fig 12-27

H19 promoter

See Fig 12-25

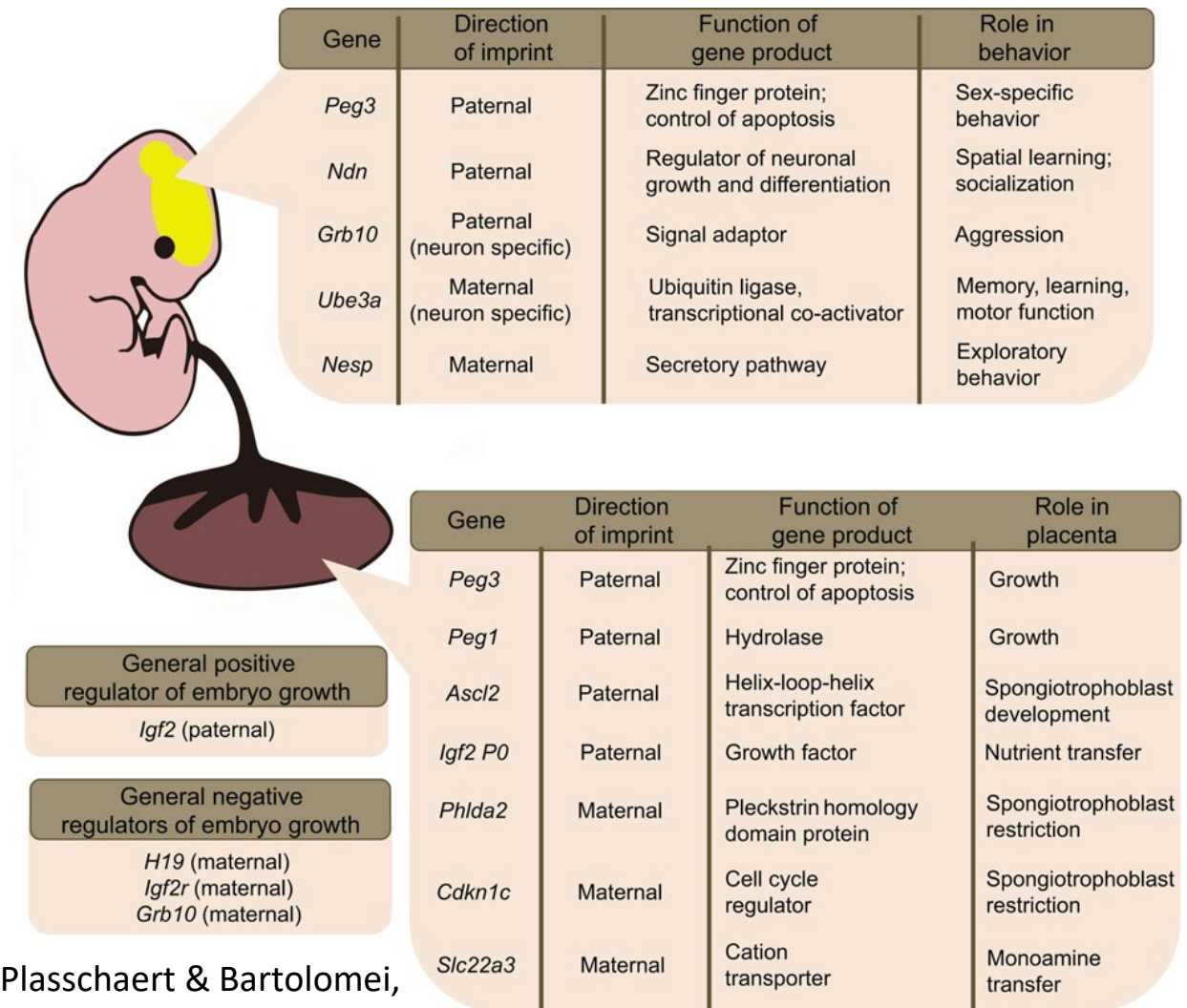
- sex-specific CpG methylation of Imprinting Control Region (ICR) – only in paternal gametes (sperm)
- unmethylated ICR binds CTCF, acts as an **enhancer-blocking insulator**
- ICR DNA methylation prevents CTCF binding; methylation in *H19* promoter silences transcription

Imprints are established during gametogenesis



Although few, imprinted genes play important roles in embryonic development

Uniparental embryos are inviable because of improper imprinting



from Plasschaert & Bartolomei, *Development* 2014;
doi:10.1242/dev.101428

REPORTS

Epigenetic Instability in ES Cells and Cloned Mice

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Hidenori Akutsu,^{3*} Konrad Hochedlinger,¹
William M. Rideout III,¹ Detlev Biniszkiewicz,¹
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Cloning by nuclear transfer (NT) is an inefficient process in which most clones die before birth and survivors often display growth abnormalities. In an effort to correlate gene expression with survival and fetal overgrowth, we have examined imprinted gene expression in both mice cloned by nuclear transfer and in the embryonic stem (ES) cell donor populations from which they were derived. The epigenetic state of the ES cell genome was found to be extremely unstable. Similarly, variation in imprinted gene expression was observed in most cloned mice, even in those derived from ES cells of the same subclone. Many of the animals survived to adulthood despite widespread gene dysregulation, indicating that mammalian development may be rather tolerant to epigenetic aberrations of the genome. These data imply that even apparently normal cloned animals may have subtle abnormalities in gene expression.

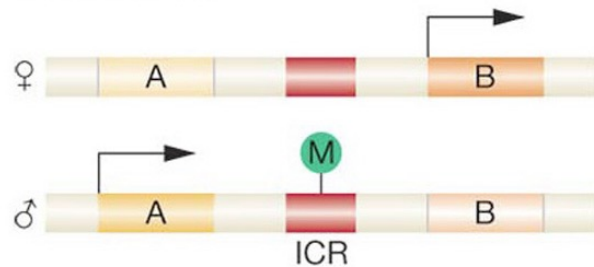
www.sciencemag.org SCIENCE VOL 293 6 JULY 2001



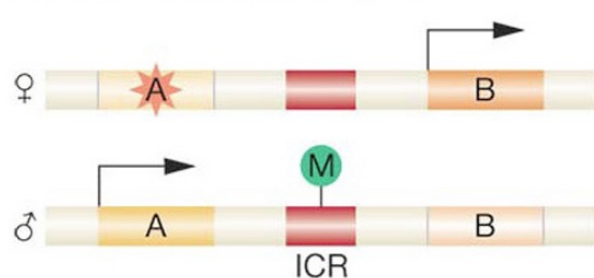
Most cloned animals don't live as long as Dolly. © SPL

Imprinting and disease inheritance

No mutations

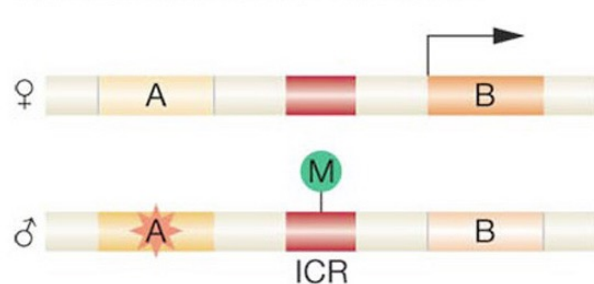


Mutation in imprinted gene



OUTCOME UNAFFECTED

Mutation in nonimprinted gene



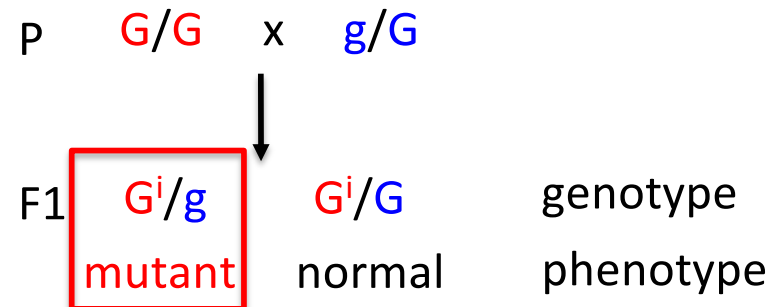
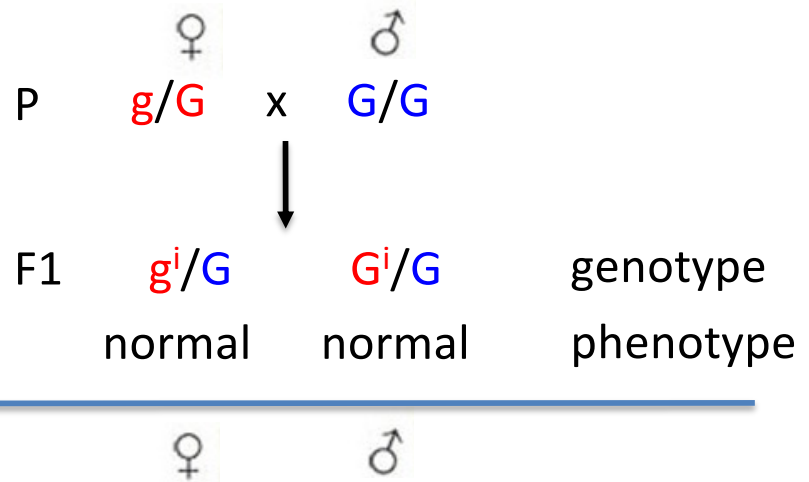
OUTCOME AFFECTED

11th Edition Fig 12-28

G = Maternally Imprinted gene

Red: maternal

Blue: paternal



Mutation in non-imprinted allele acts dominantly, depending on which parent passes it on

Which one of the following will not lead to a disorder associated with an imprinted gene?

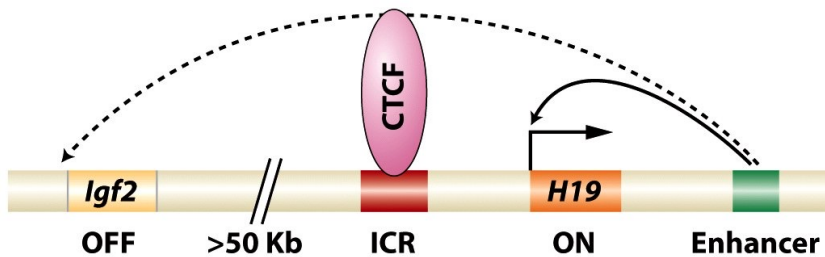
- A. mutations affecting the non-imprinted copy
- B. epimutations (defects in histone tail modification or DNA methylation that affect gene expression).
- C. uniparental disomy (Robertsonian carrier, non-disjunction, somatic cell recombination).
- D. all three mechanisms could affect imprinting

Imprinting and disease – faulty imprinting

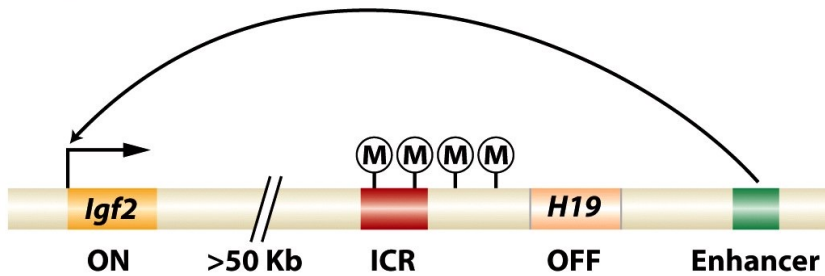
Silver-Russell syndrome - dwarfism

unaffected individual

♀ Maternal allele

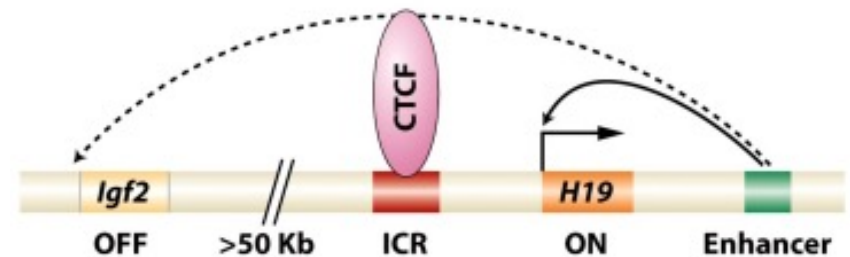


♂ Paternal allele

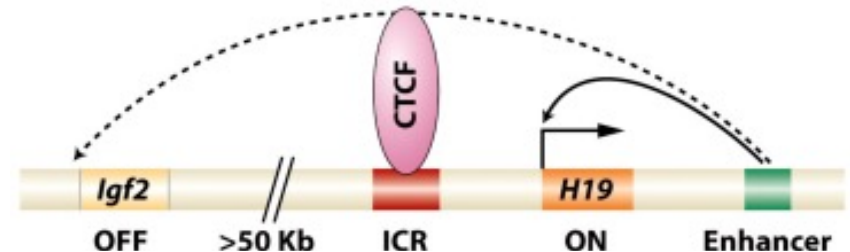


affected individual

♀ Maternal allele



♂ Paternal allele



- ~50% of cases - hypomethylation of paternal allele (“**epimutation**”) leads to repression of paternal Igf2 expression